

Accept, Transfer, Avoid or Mitigate? An Analysis of Risk Strategies Within a Network Packet Capture Simulation

JOSHUA SOURBEER, University of Central Florida, USA

Abstract—There is limited research which considers the effectiveness of risk management strategies (accept, transfer, mitigate, avoid) in a simulation environment, informed by real-world benign and malicious packet captures (PCAPs). In this study, the effectiveness of the risk management strategies will be measured and compared concerning risk escalation, threat likelihood, impact and residual risk using scenarios derived from the Open Source Foundation for Web Security (OWASP) top 10 list.

CCS Concepts: • **Software and its engineering** → **Risk management**; • **Networks** → **Network simulations**; **Network experimentation**; **Network performance modeling**; **Network performance analysis**.

Additional Key Words and Phrases: Network Simulation, PCAP Analysis, Risk Management

ACM Reference Format:

Joshua Sourbeer. 2026. Accept, Transfer, Avoid or Mitigate? An Analysis of Risk Strategies Within a Network Packet Capture Simulation. *J. ACM*, (2026), 14 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

The evolution of the modern internet, coupled with the growing complexity of networks and societal dependence on emerging technologies has shifted the cybersecurity industry away from individual operators fixing problems and towards enterprise level challenges. These challenges are further complicated through the advent of highly sophisticated cybersecurity attacks, threats and vulnerabilities. [2] Likewise, traditional defense methods such as perimeter focused security and compliance based frameworks are no longer sufficient to deal with new emerging threats. In the modern landscape of machine learning (ML) and artificial intelligence (AI) enabled threats, legacy signature and rule based defense systems can no longer be relied upon exclusively. While these advanced threats are often thought of as the most severe of modern risks, humans pose the most common threat, with 74 percent of all breaches attributed to human error, phishing or stolen credentials. [7]

In managing this newly evolving threat landscape and human enabled threats, the Risk Management Framework (RMF) continues to provide IT professionals with a stable, reliable structure for responding to new threats. [1] The risk management framework provides for a multi-step process to frame, assess, respond and monitor risk, challenging leaders to look beyond individual decisions and towards a full-scale risk focused governance strategy. [14] The foundation of this RMF informed strategy lies in the selection of an approach informed by organizational risk tolerance and objectives; acceptance, transference, avoidance or mitigation. Accepting a risk involves acknowledging that a risk is within the risk tolerance of an organization and taking no action to prevent or avoid it. Risk transference involves displacing risk to another party; for example s an insurance or cloud provider. Avoiding a risk involves completely eliminating the possibility of a

Author's Contact Information: Joshua Sourbeer, jo501433@ucf.edu, University of Central Florida, Orlando, Florida, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 1557-735X/2026/-ART

<https://doi.org/XXXXXXX.XXXXXXX>

risk, such as disconnecting a system from a network. Risk mitigation involves taking an action to reduce a risk, such as patching vulnerable systems. [14]

While the RMF provides for a reliable method to approach risk, there is a shortage of research into the empirical effects of the four approaches in a network simulation environment. Modeling and Simulation tools are frequently used for network testing due to their scalability, flexibility, and capacity to provide detailed feedback in a safe environment. However, they are often limited by a highly technical focus which detracts from organizational practice and enterprise risk management goals. [4]

Additional limitations are imposed by the tools themselves, with many viewed as patchwork solutions incapable of generating dynamic scenarios required for full spectrum risk assessments. Likewise, a competency gap is emerging where niche tools with steep learning curves are used primarily by technical analysts, but not by risk managers and decision-makers. [6]

Organizational hesitancy to employ network simulation tools is often hastened by a concern for maintaining confidentiality, availability and privacy. Bringing down a production network to run a simulation is generally not sought after due to a reduction in availability. Likewise, there are privacy and confidentiality concerns with testing live networks; packet captures (PCAPs) for example contain a great deal of identifiable data on subjects. [13]

Hesitancy to employ tools is further compounded by a widening gap between analysts and risk managers regarding defensive capabilities and strategies. This gap can be attributed to highly technical detection systems, vague machine learning algorithms and communication failures between professionals. [3] Ultimately, there is a need to shift focus away from a technical, event-based approach to cybersecurity and towards an enterprise level strategy for framing risks and selecting an appropriate risk response.

2 Background

2.1 Research Question

This research seeks to determine to what extent different risk management strategies (accept, transfer, avoid, mitigate) affect the exposure factor, likelihood, impact and residual risk of networks given benign and malicious network traffic in a simulation environment.

2.2 Risk Management Decision Making Components

For risk managers, crafting a organizational security strategy requires a full comprehension of variables which affect threat posture. Risk management is an ever-changing, iterative process where the costs and provisions of technology must be carefully balanced against the threat of organizational failure. The sections below serve to define core metrics used in RMF needed to make effective and informed decisions. [14]

2.2.1 Risk, Likelihood and Impact. Risk is defined as the effect of uncertainty on objectives; in an operational setting it represents the measure to which a component is threatened by a potential circumstance. Two key variables comprise risk; the first are adverse impacts which arise from the occurrence of a situation; impact is often measured in terms of financial loss, reputation damage, regulatory fines or disruptions. Likelihood is the second variable; it involves the probability that a certain risk or threat will occur, and is often measured via the annual rate of occurrence (ARO) which represents the number of incidents per year.[9]

2.2.2 Exposure Factor and Residual Risk. While likelihood and impact provide for a top-level view of risk, the exposure factor and residual risk offer a granular approach to analyze vulnerable systems. An organizational exposure factor is related to the specific percentage of an asset prone to any given

source. Risk managers who seek to reduce their threat surface often examine the exposure factors of different devices, network segments or vulnerable systems. Residual risk is defined as the level of remaining risk after controls and responses have been implemented in reaction to a specific event. While a low level of residual risk is optimal, no single decision can entirely eliminate uncertainty; the goal of RMF is to reduce the residual risk to an acceptable level, with an organization's risk tolerance. [14]

2.3 Challenges in Selecting a Risk Strategy

Responding to residual risk and emerging threats requires to select an appropriate risk strategy to accept, transfer, avoid or respond to risk, informed directly by likelihood, impact and the exposure factor. Current approaches to selecting a valid strategy are hindered by the lack of a cohesive perspective regarding how technical knowledge should inform organizational practice. Specifically, this arises primarily due to a localization of decision-making within operations centers rather than being driven by executive management, policies and stakeholders. This thought process reinforces an invalid assumption that network security data is only relevant for technical tasks, isolating it from other enterprise wide uses.[1]

Furthermore, the selection of a risk strategy can be difficult to apply across domains, often complicated by network complexities where segments of a network are created and configured at different times by various parties. [26] Accessibility and usability inhibitors also hinder strategic decision-making; many network analysis tools require command input with manual queries which are inaccessible to managers, resulting in a disconnect between insights and policy creation. Across the industry, an increasing level of attention is focusing on decreasing the time to arrive at these insights. However these efforts often result in stopping at discovering new vulnerabilities rather than translating insights into RMF strategies. [10][21]

Additional technical challenges arise when analyzing large volumes of network data to make informed decisions; false negatives and hallucinations can have an extremely detrimental effect identifying and visualizing the threat surface. [20] Likewise, there are specialized threats such as USB traffic and lateral movements which necessitate advanced reasoning that traditional metrics simply fail to capture when building strategies. [19] [23]

Finally, selecting a risk strategy is also made difficult via a growing communication disconnect between analysts and decision makers, often accelerated via differences in focus. While analysts tend to judge success through metrics such as uptime, connectivity and incidents, executive management prefers to examine return on investment, capital gains and losses. [1]

2.4 Using Packet Captures and Simulation to Drive Risk Strategy Evaluation

Resulting from these challenges divide current approaches and research between advanced network analysis frameworks, network simulation, and threat focused risk management. Currently, these approaches are extremely domain specific and rarely make connections between technical variables and risk reduction strategies. As a result, there is a significant lack of research into how various risk strategies perform under different real-world scenarios, both benign and malicious.[4][1]

2.4.1 Advanced Network Analysis Frameworks. A variety of cutting edge frameworks exist for analyzing network traffic using large language models and ensemble based models. [25] These models are capable of ingesting large volumes of packet capture data, and they have been shown to be extremely effective in detecting ongoing attacks such as distributed denial of service (DDOS) in real-time. [12] Similarly, advanced models such as Llama III are capable of correlating packet captures and logs in order to further analyze complex threats. [5] However, despite these rapid advancements in analysis there remains a significant gap between machine learning enabled analysis

tools and communication networks. Similarly, the complexity of these tools, the learning curve and the vagueness of their processes often deters those without extensive technical knowledge from using them. [11] In addition, relying on machine learning frameworks exclusively for senior decision making can be problematic when policy makers lack the required knowledge to understand the capabilities and limitations of these strategies. [3]

2.4.2 Network Simulation. Network simulation environments can provide valuable testing grounds for analyzing the effectiveness of different risk strategies. These environments can provide for realistic portrayal of communications protocols and messaging, while offering customizable configuration and testing. A review of six popular frameworks, including CORE, Cisco Packet Tracer and ns-3 revealed that while each offered a realistic, high fidelity testing ground, organizational objectives gave way to technical objectives. [22]

In addition, there remains a need to incorporate live network traffic data into simulations. Some approaches attempt to develop parallel discrete event simulations to model highly specific networks, such as underwater acoustics. [18] Other approaches seek to employ mirroring live and cloned packet capture data using virtualized network devices. Including real-world network traffic in simulation environments is both useful and critical to providing for a realistic, and comprehensive evaluations without risking violations of Service Level Agreements (SLA) or network policies. [24]

2.4.3 Threat Focused Risk Management. An abundance of research examines how packet captures can provide valuable insights into attack vectors that are not inherently obvious, such as covert channels. [27] Packet captures can also reveal whether ongoing cybersecurity attacks are occurring, validating their use as an accurate snapshot of security risks that may not appear in log files. [16] Likewise, additional research found a series of risks inherent to digital certificates that could only be discovered via certificate transparency logs. [17] To complement gaps found in limiting analysis to packet captures, further research was done to combine network and log streams into a singular framework, which was found to be highly capable against irregular attacks. [15]

However, a critical limitation of these noted studies is their reliance and focus on attackers and threats rather than an enterprise focus. In treating threat discovery, as the final step in the process rather than the input, current methodologies fail to translate findings into managerial quantifiable metrics such as exposure factors, likelihood and impact. [8]

Finally, a consistent shortcoming within threat focused frameworks is an inaccurate level of realism in scenarios. Most approaches rely upon threat signatures, traces or synthetic indicators rather than multi-step, narrative driven events from authoritative sources such as OWASP. [27]

2.5 Research Question and Hypothesis

The core question of the research is determine to what extent different risk management strategies (accept, transfer, avoid, mitigate) affect the exposure factor, likelihood, impact and residual risk of networks given benign and malicious network traffic in a simulation environment. It is hypothesized that a balanced approach to risk is optimal to reduce impact in benign environments, but a high-risk environment necessitates a mitigation focused approach.

3 Methodology

Rather than model an approach where a large library of detailed, procedural security controls could be applied to reducing risk, the methodology used in this research models the effects of controls, reducing the available actions to accept, transfer, avoid or mitigate. In order to compare each risk treatment strategy to one another while evaluating the exposure factor, likelihood, impact and residual risk, a new simulation tool was required to test these strategies using packet captures. PackSim is a desktop simulation tool which permits the upload of packet captures; .PCAPs are

analyzed for devices, ports and protocols, and a set of known vulnerabilities. Each device is then assigned an initial risk score, which feeds an overall starting network risk score. PackSim includes twenty scenarios from the OWASP Top 10 vulnerabilities list; scenarios are dynamically chosen for each .PCAP based on realistic outcomes and whether the threat applies to the given network. Each replication contains five generalized phases of attack exploitation; initial detection, early exploitation, privilege escalation, impact and data exposure, and full compromise. PackSim allows for the manual selection of a strategy at each phase, or an auto simulation which permits numerous replications to be conducted. In addition to the strategy, PackSim allows the selection of a severity level (low, medium or high) which dictates how often aggressively the individual or organization intends to pursue a given strategy.

This experimental plan seeks to examine risk strategy comparison, exposure and residual risk analysis in a three-part replication strategy. These trials will all occur within a single simulation framework to determine how differing risk management strategies affect cybersecurity outcomes within both benign and malicious networks. Objective 1 establishes a comparison between the strategies to evaluate their effects on likelihood and impact of threat events. Objective 2 isolates the exposure factor to determine whether differences examined in objective 1 are strategy or vulnerability derived to answer how the exposure factor affects the escalation of risk. The final objective shifts focus from technical metrics to governance by evaluating how accepting risk impacts the level of residual risk after controls are applied. In conclusion, objectives 1-3 combine to answer the hypothesis by using both benign and malicious packet capture data across all strategies to examine likelihood, impact, residual risk and risk escalation.

3.1 Research Objectives

3.1.1 Objective 1: Optimal Risk Strategy for Benign/ Malicious Networks. This objective seeks to compare risk strategies (accept, avoid, transfer, mitigate) based on how they reduce the likelihood of an occurrence and the impact of a threat, attempting to isolate the single most effective strategy in regards to low likelihood and impact levels.

3.1.2 Objective 2: Influence of the Exposure Factor on Risk Escalation. Objective two seeks to evaluate how the exposure factor contributes to overall risk escalation within all five phases of an OWASP informed simulation run. Using a balanced strategy involving a combination of all four approaches, the exposure factor will be measured during each simulation run to determine whether strategy performance differences are driven primarily by vulnerabilities or the strategy itself.

3.1.3 Objective 3: Acceptable Residual Risk Levels During Risk Acceptance. Many organizations decide to accept risk based on circumstances, financial hardships, or low likelihood levels. Objective 3 will seek to determine what acceptable levels of residual risk are when employing risk acceptance strategies of low, medium and high severity levels in both benign and malicious networks.

3.2 Experimental Setup

3.2.1 Datasets. In total, 10 real-world packet benign captures were individually selected from a wide variety of different network types, alongside 10 real-world packet captures containing malicious traffic or known vulnerabilities. The networks, threats and scenarios are described in the Table 3 below for each packet capture included in the research.

3.2.2 Experimental Flow. In each experiment, a packet capture is uploaded into PackSim which serves as the primary network in use. Upon upload, each packet capture is drawn in a network traffic diagram, then analyzed for a set of 10 network risk indicators derived from shallow packet analysis, itemized below; devices are then assigned by type and OSI Layer.

- Open Services
- Insecure/Outdated Protocols
- Remote Access
- Lateral Movement Pathways
- Abnormal Traffic Volume
- Device Role Mismatch
- Dangerous Services
- Network Signature Anomalies
- Uncertainty/Hidden Risk

Individual device risk scores and an overall network risk score are then printed to the console, assigned and normalized from 0-1 to account for various PCAP sizes and device counts, considering how many devices expose lateral movement ports, remote administration and vulnerable services. Upon initiating the simulation, an initial risk management strategy will be selected with corresponding severity level; scenarios are selected from the OWASP Top 10, with each simulation replication containing all five phases of risk response. Throughout the duration of the simulation, key performance indicators (KPIs) dynamically change based on security posture, compromise level and operational strain. Upon completion, KPIs are updated and posted to graphs for viewing and download.

3.3 Results

3.3.1 Experiment 1: Optimal Risk Strategy For Reducing Likelihood and Impact. In experiment one, each risk strategy was compared and measured in a series of 10,000 simulation replications which evaluated the likelihood of threat continuity and financial impact at the end of each trial. Risk avoidance consistently provided the highest raw scores for likelihood with a low mean of 2.45 percent likelihood in benign environments and 3.02 percent in malicious environments. However, the financial impact of risk avoidance was significantly impactful, with the average financial loss for a threat event exceeding double the amount of any strategy. This can be attributed to revenue losses from unused, air gapped or offline systems taken out of production to avoid risk scenarios. Risk mitigation yielded the lowest mean financial loss in a benign environments (5,155) while risk transference demonstrated the lowest mean impact values in malicious environments (7725). In combination, these results support the initial hypothesis that in benign environments, a balanced strategy which employs all four of the risk strategies is optimal to keep impact minimal and likelihood at a manageable rate.

3.3.2 Experiment 2: Effect of the Exposure Factor on Likelihood, Compromise and Risk Escalation. Experiment two examined the exposure factor's affect on likelihood of threat continuity, the degree to which organizations can become adversely compromised by a threat, and the level of risk increase throughout the life cycle of a security incident. A balanced approach to risk management was examined at three network volatility levels (variable, unstable and chaotic), which introduce variability for exposure and security posture. In total, 6,000 replications were conducted for across each of the volatility levels, in both benign and network environments. The mean exposure rate across all trials for experiment 2 in both environments was 5.195, a significantly lower rate for exposure than risk acceptance in trial 1 (11.5), indicating that a balanced strategy can reliably reduce the exposure rate in both environments. Additionally, the two benign and malicious chaotic simulation runs which ended with a poor security posture both contributed to the highest levels of exposure, indicating that security posture and operational risk decisions can reduce the exposure of a network even with a high initial risk score under malicious conditions.

3.3.3 Consequences of Risk Acceptance Concerning Residual Risk. In experiment 1, Risk acceptance proved to have the highest levels of residual risk (28.1 for benign and 28.7 for malicious environments). However, acceptance demonstrated other advantages such as the lowest internal organizational strain means (1.6 for benign and 2.3 for malicious) and also relatively low impact levels which beat out risk avoidance and balanced strategies in both environments. Experiment three isolates the risk acceptance strategy, examining residual risk to shift focus to governance, acknowledging that many organizations choose to accept risk simply because the costs to mitigate exceed the single loss expectancy (SLE) for a threat event. After 6,000 replications, risk acceptance was compared across all 20 benign and malicious PCAPs, in both variable, unstable and chaotic network conditions. Lower asset value and residual risk scores generally were aligned with lower impact scores. However, increasing the volatility of the network did not result in higher financial losses consistently, indicating that risk acceptance can provide advantages and cost minimization given the right set of circumstances.

3.4 Discussion

3.4.1 Research Objective 1: Optimal Strategy in Benign and Malicious Environments. The simulation results conclude that in both benign and malicious environments, a combination of multiple risk management approaches is recommended to minimize the financial impact of a threat and the likelihood of the threat continuing. As supported by experiment 1, 10,000 simulation replications indicate that in benign environments, risk avoidance and mitigation had the lowest likelihood/impact scores, while risk mitigation and transference had the lowest scores for likelihood/impact respectively.

3.4.2 Research Objective 2: Influence of the Exposure Factor. Experiment 2 sought to answer when networks are exposed, how does the exposure factor contribute to risk escalation throughout the security event life cycle. From the results, the network exposure score rises with the volatility level in the simulation, but the total risk and residual risk levels remain unaffected substantially by higher exposure rates. In addition, malicious rates often have lower exposure but higher likelihood rates. These observations serve to support the viewpoint that the exposure factor shapes the likelihood and visibility of risk, but the value itself does not drive impact or total risk. Likewise, the security posture had a significant affect on the exposure factor, driving EF values down even in malicious scenarios. This aligns with research that concludes that the exposure factor is a situational risk-surface index, not a direct measurement of impact or compromise level. It is a supplemental, informative metric designed to provide a snapshot into current security levels rather than a predictive indicator of losses.

3.4.3 Research Objective 3: Residual Risk During Risk Acceptance.

3.4.4 Partial Acceptance of the Original Hypothesis.

3.4.5 Environment Assumptions.

3.4.6 Known Weaknesses and Limitations. A significant limitation of the PackSim framework is that it is designed only to provide shallow packet analysis from PCAPs, examining primarily packet headers rather than payloads which limits the variety of the initial device and network risk indicators.

3.5 Conclusion

3.5.1 Contributions. The research outcomes examined in this paper filled a shortage of literature concerning risk management strategies in a simulation environment, informed by real-world packet

captures (PCAPs). PackSim, the python-based simulation framework developed for this project is completely open-source, licensed under the GNU GPL v3 License and available for download, modification and experimentation.

3.5.2 Future Work and Research Extensions. Future work with the PackSim framework will consist of future updates to the simulation back-end framew

3.6 References

References

- [1] Scott Ainslie, Dean Thompson, Sean Maynard, and Atif Ahmad. 2023. Cyber-threat Intelligence for Security Decision-making: A review and research agenda for practice. *Computers and Security* 132 (Sep 2023), 103352. doi:10.1016/j.cose.2023.103352
- [2] Ayesha Ajmal, Maryam Doborjeh, and Jairo Gutierrez. 2025. Detection of fileless malware through network traffic analysis. *2025 IEEE 35th International Telecommunication Networks and Applications Conference (ITNAC)* (Nov 2025), 1–4. doi:10.1109/itnac66378.2025.11302628
- [3] Omar Alshaikh, Simon Parkinson, and Saad Khan. 2023. *Exploring perceptions of decision-makers and specialists in defensive machine learning cybersecurity applications: The need for a standardised approach* (2023). doi:10.2139/ssrn.4582920
- [4] Uchenna Daniel Ani, Mohammed Al-Mhiqani, Nilufer Tuptuk, Stephen Hailes, and Jeremy Daniel McKendrick Watson. 2025. Socio-technical security modelling and simulations in cyber-physical systems: Outlook on knowledge, perceptions, practices, enablers, and barriers. *IET Cyber-Physical Systems: Theory and Applications* 10, 1 (Jan 2025). doi:10.1049/cps2.70017
- [5] Hyun-Min Choi and Youngseok Lee. 2025. Fine-tuning LLAMA3 for integrated analysis of Network Packet and System Log Data. *2025 30th Asia-Pacific Conference on Communications (APCC)* (Nov 2025), 1–6. doi:10.23919/apcc64555.2025.11279921
- [6] Manal Ghanem, Nicholas Micallef, Sara Correia-Hopkins, and Jonathan-Lee Jones. 2025. Barriers and challenges of network simulation tools: A case study on core/eman. *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering* (2025), 397–417. doi:10.1007/978-3-031-87345-4_26
- [7] Gazi Mohammad Haque, Dhiraj Kumar Akula, Yaseen Shareef Mohammed, Asif Syed, and Yeasin Ararat. 2025. Cybersecurity risk management in the age of Digital Transformation: A Systematic Literature Review. *The American Journal of Engineering and Technology* 7, 08 (Aug 2025), 126–150. doi:10.37547/tajet/volume07issue08-14
- [8] Soumyadeep Hore, Jalal Ghadermazi, Diwas Paudel, Ankit Shah, Tapas Das, and Nathaniel Bastian. 2025. Deep Packgen: A deep reinforcement learning framework for Adversarial Network Packet Generation. *ACM Transactions on Privacy and Security* 28, 2 (Feb 2025), 1–33. doi:10.1145/3712307
- [9] International Standards Organization. [n. d.]. ISO 31000:2018. <https://www.iso.org/standard/65694.html>
- [10] Sridhar Iyer, Kunal Joshi, Jharana Solanki, Rachita Samant, and Jay Patel. 2023. Packetenizer - modern PCAP analyzer. *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)* (Jul 2023), 1–7. doi:10.1109/iccncnt56998.2023.10307132
- [11] Marc Katzef, Andrew C. Cullen, Tansu Alpcan, Christopher Leckie, and Justin Kopacz. 2022. Wireless network simulation to create Machine Learning Benchmark Data. *GLOBECOM 2022 - 2022 IEEE Global Communications Conference* (Dec 2022), 6378–6383. doi:10.1109/globecom48099.2022.10000685
- [12] Farheen Khanum, P Sree Lakshmi, and Harsha Vardhan Reddy. 2024. Network analysis of Multisource Packet Capture files using machine learning. *2024 5th International Conference on Circuits, Control, Communication and Computing (I4C)* (Oct 2024), 119–124. doi:10.1109/i4c62240.2024.10748490
- [13] Michele Mastroianni, Francesco Palmieri, Massimo Ficco, Rafał Kozik, and Michał Choraś. 2023. Privacy risk analysis and metrics in capturing and storing network traffic. *2023 24th International Conference on Control Systems and Computer Science (CSCS)* (May 2023), 580–585. doi:10.1109/cscs59211.2023.00097
- [14] National Institute of Standards and Technology. 2012. Managing Information Security Risk: The Risk Management Framework. Retrieved February 4, 2026 from <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-39.pdf> NIST Special Publication 800-39.
- [15] Nadia Niknami, Vahid Mahzoon, Rajorshi Biswas, Slobadan Vucetic, and Jie Wu. 2025. An interpretable multi-modal transformer-based intrusion detection system utilizing log messages and PCAP files. *2025 IEEE 22nd International Conference on Mobile Ad-Hoc and Smart Systems (MASS)* (Oct 2025), 271–279. doi:10.1109/mass66014.2025.00048
- [16] Vasudeva Pai, Abhishek S. Rao, Krishna Prasad D S, Devidas, and Karthik Pai B H. 2022. Analysis of routing protocol for low power and lossy networks attack behavior on the internet of things. *2022 International Conference on Artificial Intelligence and Data Engineering (AIDE)* (Dec 2022), 52–57. doi:10.1109/aide57180.2022.10060749

- [17] Damiano Ravalico, Martino Trevisan, and Idilio Drago. 2025. Poster—reconnaissance via certificate transparency logs: Exposing self-hosted web applications. *Proceedings of the 21st International Conference on emerging Networking EXperiments and Technologies* (Nov 2025), 34–35. doi:10.1145/3765515.3771748
- [18] Xun Sun, Guoliang Weng, and Lizhao You. 2025. Poster: Towards efficient and faithful underwater acoustic network simulation with Direct Code Execution. *Proceedings of the 21st International Conference on emerging Networking EXperiments and Technologies* (Nov 2025), 36–38. doi:10.1145/3765515.3771751
- [19] Marian Ticu. 2021. USB traffic analyzer - digusb. *2021 12th International Symposium on Advanced Topics in Electrical Engineering (ATEE)* (Mar 2021), 1–5. doi:10.1109/atee52255.2021.9425263
- [20] Lukasz Tulczyjew, Ihor Biruk, Murat Bilgic, Charles Abondo, and Nathanael Weill. 2024. PCAPVision: PCAP-based high-velocity and large-volume network failure detection. *Proceedings of the 2024 SIGCOMM Workshop on Networks for AI Computing* (Aug 2024), 26–33. doi:10.1145/3672198.3673796
- [21] Juraj Uhlar, Martin Holko, and Vit Rusnak. 2021. PCAPFunnel: A tool for rapid exploration of packet capture files. *2021 25th International Conference Information Visualisation (IV)* (Jul 2021), 69–76. doi:10.1109/iv53921.2021.00021
- [22] Theodore A. V, Eric Gamess, and David Thornton. 2021. A survey of wireless network simulation and/or emulation software for use in higher education. *Proceedings of the 2021 ACM Southeast Conference* (Apr 2021), 63–70. doi:10.1145/3409334.3452066
- [23] Louis Van Langendonck, Ismael Castell-Uroz, and Pere Barlet-Ros. 2024. Towards a graph-based foundation model for Network Traffic Analysis. *Proceedings of the 3rd GNet Workshop on Graph Neural Networking Workshop* (Dec 2024), 41–45. doi:10.1145/3694811.3697817
- [24] Francisco Germano Vogt, Zhiheng Yang, Victor Hugo Lopes, Fabricio Rodriguez, Marcelo Caggiani Luizelli, Christian Esteve Rothenberg, and Chrysa Papagianni. 2025. Poster: QueuePilot: Towards adaptive queue scheduling via Live Digital Twins. *Proceedings of the 21st International Conference on emerging Networking EXperiments and Technologies* (Nov 2025), 39–40. doi:10.1145/3765515.3771755
- [25] Francisco Germano Vogt, Zhiheng Yang, Victor Hugo Lopes, Fabricio Rodríguez, Marcelo Caggiani Luizelli, Christian Esteve Rothenberg, and Chrysa Papagianni. 2025. Poster: Reconstructing LLM training workloads: A topology- and model-aware network tester. *Proceedings of the 21st International Conference on emerging Networking EXperiments and Technologies* (Nov 2025), 41–42. doi:10.1145/3765515.3771757
- [26] Lukas Wüsteney, David Hellmanns, Markus Schramm, Lukas Osswald, René Hummen, Michael Menth, and Tobias Heer. 2022. Analyzing and modeling the latency and jitter behavior of mixed industrial TSN and DetNet Networks. 91–109 pages. doi:10.1145/3555050.3569138
- [27] Marco Zuppelli and Luca Caviglione. 2021. PcapStego: A tool for generating traffic traces for experimenting with network covert channels. *Proceedings of the 16th International Conference on Availability, Reliability and Security* (Aug 2021), 1–8. doi:10.1145/3465481.3470067

3.7 Appendix A: PackSim Supplemental Formulas and Tables

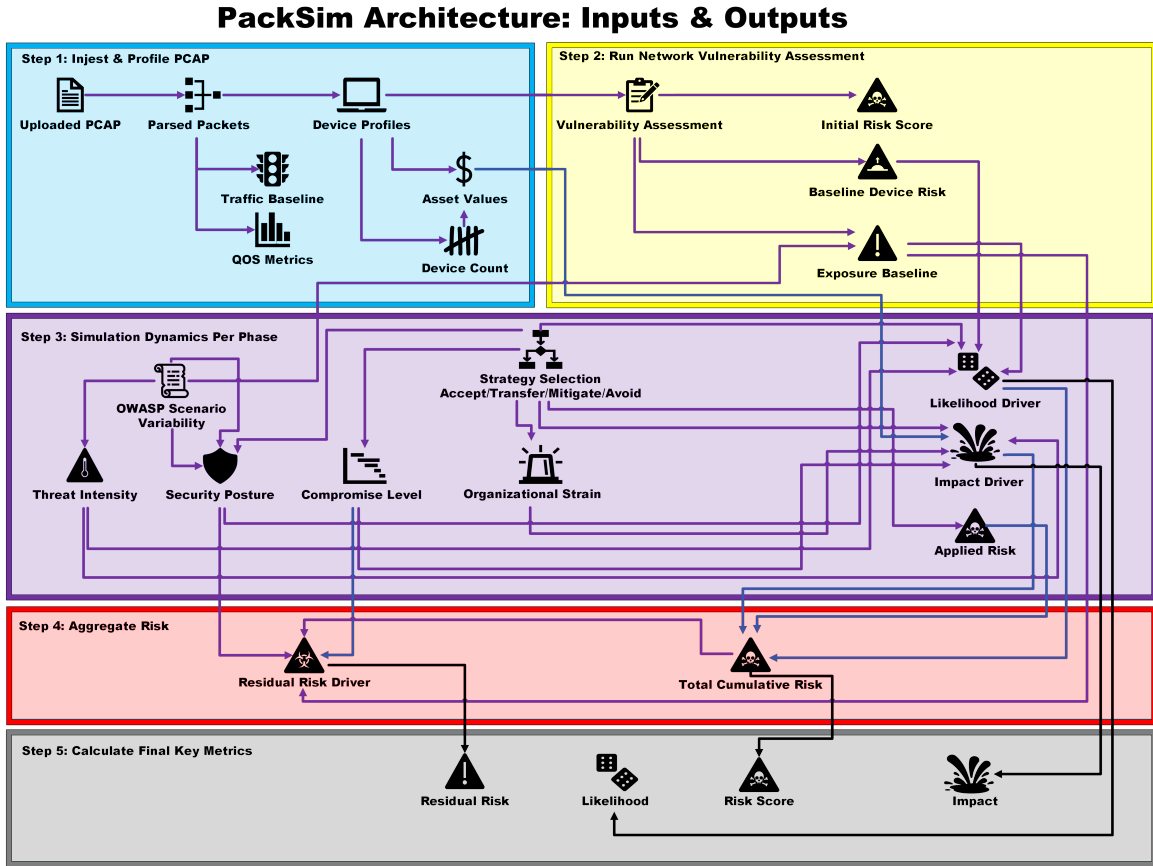


Fig. 1. PackSim Simulation Architecture

Metric	Formula	Inputs	Outputs
Risk (Per Phase): Total Risk Accumulated During Simulation Run	$R_{\text{phase}} = B \times A$	B : Threat base risk A : Action multiplier	Percent contribution to cumulative risk
Likelihood: The Probability of the Current Threat Continuing Beyond the Simulation Run	$\frac{1}{1 + e^{-2(R_{\text{eff}} - 0.6)}}$ $R_{\text{eff}} = L_0 \times A \times (1 + E) \times \frac{1}{P} \times (1 + C)$	L_0 : Base device risk A : Action multiplier E : Exposure P : Security posture C : Compromise level	Probability of threat progression
Impact: Financial (USD) Estimation of the Single Loss Expectancy (SLE) for a Security Incident	$V(1 + S)(1 + 1.5C)$	V : Asset value S : Operational strain C : Compromise level	Impact score and estimated financial loss
Residual Risk: Measurement of How Much Risk Remains in the System After a Simulation Run	$R_{\text{total,new}} = R_{\text{total,old}} + R_{\text{phase}} \left(1 - \frac{R_{\text{total,old}}}{100} \right)$ $S_P = \frac{P - 0.5}{1.5}$ $C = 0.5S_P + 0.3(1 - E) + 0.2(1 - C_I)$ $R_{\text{res}} = R_{\text{total}}(1 - C)$	R_{phase} : Phase risk percent R_{total} : Cumulative risk P : Security posture (0.5–2.0) E : Exposure (0–1) C_I : Compromise level (0–1) C : Control effectiveness	Post-control cumulative risk percentage
Exposure: Dynamic Measurement of How Exposed the Network is at Any Given Time	$E = \text{clip}(E_{\text{raw}}, 0.05, 0.75)$ $E_{\text{raw}} = R_N(1 + 0.2C) \frac{1}{P} \times 0.7$	R_N : Normalized network risk C : Compromise level P : Security posture	Dynamic exposure level (5%–75%)

Table 1. PackSim’s Core Metrics and Formulas

PCAP Index	Type	Background	Device Count	Initial PackSim Risk Level
1 (Smith–Hart)	Benign	Industrial Control System	3	8
2 (Smith–Tridium 1)	Benign	Industrial Control System	7	23
3 (Smith–Tridium 2)	Benign	Industrial Control System	4	0
4 (KargsNet–GetEvent)	Benign	Home Office	2	0
5 (KargsNet–SpecialEvent)	Benign	Home Office	5	4
6 (Ultimate PCAP 1)	Benign	Compilation	3	2
7 (Ultimate PCAP 2)	Benign	Compilation	4	12
8 (Ultimate PCAP 3)	Benign	Compilation	5	10
9 (Westermo)	Benign	Industrial Control System	8	46
10 (Wireshark Samples)	Benign	Industrial Control System	15	2
11 (MTA Net)	Malicious	Bazar Loader Malware	52	28
12	Malicious	Bumblebee Loader Malware	94	61
13	Malicious	IcedID Banking Trojan	5	7
14	Malicious	Pikabot Malware Trojan	7	Medium
15	Malicious	Bandook Remote Access Trojan (RAT)	14	High
16	Malicious	Darkgate Remote Access Trojan (RAT)	11	High
17	Malicious	Latrodectus Malware	9	High
18	Malicious	Remcos Remote Access Trojan (RAT)	59	23
19	Malicious	xWorm Remote Access Trojan (RAT)	5	9
20	Malicious	LummaStealer Malware	44	35

Table 2. PCAP Dataset Overview with Initial PackSim Risk Levels

Component	Formula/Variable	Inputs (Source)	Outputs
Base Device Risk	$L_0 = \sum s_i + 1_{(S \geq 15)} \lfloor 2 \ln(S) \rfloor$	s_i : PCAP Risk Indicators S : Raw cumulative device risk score	Device baseline risk (log-dampened)
Action Multiplier	ACCEPT = 1.00 MITIGATE = 0.40 TRANSFER = 0.60 AVOID = 0.10	Defender strategy selection from manual or auto-sim	Phase risk scaling factor
Security Posture	$P_{new} = \text{clip}(P_{old} + \Delta P, 0.5, 2.0)$	ΔP : Action-based adjustment by phase (+0.05 mitigate, 0.03 accept, etc.)	Bounded defensive posture
Compromise Level	$C_{new} = \text{clip}(C_{old} + \Delta C, 0, 1)$	ΔC : Phase risk dependent gain	Bounded compromise state (0–1)
Asset Value	V	Device profile assignment (default = 50)	Impact base weight
Operational Strain	S	Model state variable	Strain multiplier in impact
Applied Phase Risk	$R_{\text{phase}} = B \times A$	B : Threat base risk A : Action multiplier	Phase risk contribution
Cumulative Risk	$R_{new} = R_{old} + R_{\text{phase}} \left(1 - \frac{R_{old}}{100}\right)$	Previous risk state and phase risk	Saturating total risk (0–100)
Normalized Network Risk	$R_n = \frac{R_{network}}{N \times 20}$	Aggregated device risk N : Device count 20: Max assumed per-device risk	Baseline exposure driver (0–1)
Exposure (Raw)	$E_{raw} = R_n (1 + 0.2C) \frac{1}{P} \times 0.7$	R_n : Normalized network risk C : Compromise level P : Security posture	Unbounded exposure value
Exposure (Clipped)	$E = \text{clip}(E_{raw}, 0.05, 0.75)$	Raw exposure	Bounded exposure (5%–75%)
Effective Risk Construction	$R_{\text{eff}} = L_0 \times A \times (1 + E) \times \frac{1}{P} \times (1 + C)$	Combined state variables	Multiplicative amplification
Likelihood Transformation	$L = \frac{1}{1 + e^{-2(R_{\text{eff}} - 0.6)}}$	R_{eff}	Probability (logistic S-curve)
Impact Escalation	$I = V (1 + S) (1 + 1.5C)$	V : Asset value S : Strain C : Compromise	Escalated impact score
Total Assets	N	Unique device profiles from PCAP	Network size

Table 3. Supporting Risk Model Components and Dynamic Transformations

3.8 Appendix B: Experiment 1 Tables and Graphs

Strategy	Total Risk	Risk Change	Likelihood	Impact	Residual Risk
ACCEPT-BENIGN	66.4	+55.7	26.89	(\$5,589.90)	28.1
MITIGATE-BENIGN	33.6	+22.9	9.85	(\$5,155)	14.1
TRANSFER-BENIGN	46.3	+35.6	14.96	(\$5,175)	19.4
AVOID-BENIGN	9.7	-1.0	2.45	(\$13,635)	4
BALANCED-BENIGN	42.9	+32.2	13.68	(\$7,530)	18.2
ACCEPT-MALICIOUS	68.1	+39.7	37.87	(\$8,550)	28.7
MITIGATE-MALICIOUS	35	+6.6	11.82	(\$7,960)	14.8
TRANSFER-MALICIOUS	48.2	+19.8	18.74	(\$7,725)	20.3
AVOID-MALICIOUS	10	-18.4	3.02	(\$22,605)	4.3
BALANCED-MALICIOUS	44.6	+16.2	17.4	(\$10,575)	18.8

Table 4. PackSim Strategy Outcomes Across Benign and Malicious Network Environments

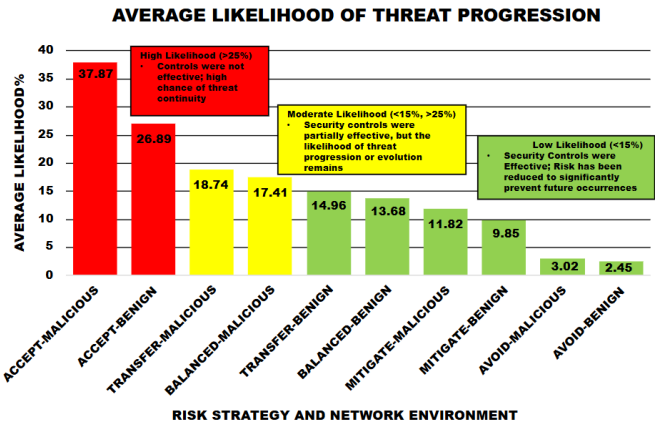


Fig. 2. Average Threat Likelihood by Risk Strategy Across Benign and Malicious Network Environments

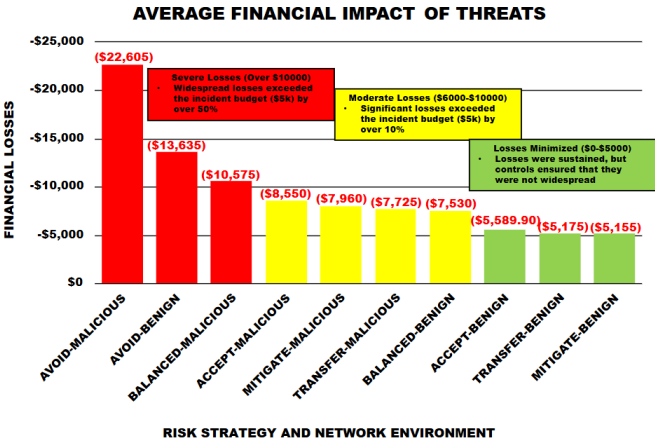


Fig. 3. Average Threat Likelihood by Risk Strategy Across Benign and Malicious Network Environments

3.9 Appendix C: Experiment 2 Tables and Graphs

Strategy	Environment	Volatility	Replications	Exposure	Risk +/-	Strain	Compromise	Posture
Balanced	Benign	Variable	1000	4.83	+32.1	4.6	6	76.6
Balanced	Benign	Unstable	1000	5.48	+32.2	4.4	6.4	76.8
Balanced	Benign	Chaotic	1000	6.82	+32	4.5	6.3	60.1
Balanced	Malicious	Variable	1000	3.82	+15.8	5.4	6.2	78
Balanced	Malicious	Unstable	1000	3.84	+16.9	5.2	6.7	76.8
Balanced	Malicious	Chaotic	1000	6.38	+15.5	5.1	6.7	54.1

Table 5. Balanced Strategy Average Exposure, Risk Escalation, Organizational Strain, Compromise Levels, and Security Posture

3.10 Appendix D: Experiment 3 Tables and Graphs

Strategy	Environment	Volatility	Replications	Asset Value	Start Risk	Total Risk	Residual	Impact
Accept	Benign	Variable	1000	28.4	10.7	42.8	15	-\$7,515
Accept	Benign	Unstable	1000	28.4	10.7	42.9	14.9	-\$7,245
Accept	Benign	Chaotic	1000	28.4	10.7	42.7	16.9	-\$7,405
Accept	Malicious	Variable	1000	46.9	28.4	44.2	15.5	-\$11,385
Accept	Malicious	Unstable	1000	46.9	28.4	45.3	15.7	-\$10,740
Accept	Malicious	Chaotic	1000	46.9	28.4	43.9	17.8	-\$10,605

Table 6. Accept Strategy Risk Metrics Across Environmental Conditions